

Halfbrick Studios: Slicing into the gaming market with Google Cloud

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ABOUT HALFBRICK

With Google Cloud, Halfbrick Studios is accelerating time to market, reducing costs, and enhancing its ability to achieve its long-term retention metrics. The business is positioned itself for growth through new game launches during the current boom in the up of mobile gaming.



About Halfbrick Studios

Founded in 2001, Halfbrick Studios has expanded from developing licensed titles to releasing a portfolio of games on multiple platforms. The success of Fruit Ninja on iPhone and iPad has powered the studio to become a globally recognized independent game developer.

Industries: Gaming

Google Cloud results

- Cuts the cost of processing analytics and event data from AUD\$24 per day to AUD\$4 per day
- Supports long-term objective of retaining 5% of users 180 days after they install a Halfbrick Studios game
- Eliminates the need to recruit an engineer to manage infrastructure

Location: Australia

Reduced
time to
new game
production
three times
one month

Google Cloud (<https://cloud.google.com/>)

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Cloud Run (<https://cloud.google.com/run/>)

Cloud Firestore

(<https://cloud.google.com/firestore/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Firebase (<https://firebase.google.com/>)

With the coronavirus pandemic prompting governments to limit nonessential travel and restrict social gatherings, mobile gaming is surging in popularity. Analysts IDC and App Annie recently

released

(<https://venturebeat.com/2020/05/27/app-annie-mobile-gaming-will-widen-its-lead-over-console-and-pc-titles-in-2020/>)

a report stating users downloaded 30% more mobile games per week in April 2020 than in January 2020, while consumer spending on mobile gaming was poised to outstrip spending on PC gaming by 2.8x and on home game consoles by 3.1x. Brisbane, Australia-headquartered gaming business Halfbrick Studios is preparing to surf this wave of demand with the release of Fruit Ninja 2—the second iteration of its highly popular mobile game.

Founded in 2001 as a gaming business commissioned to create titles for publishers, Halfbrick Studios started building its own IP in 2007/08 when Apple introduced the iPhone. The fledgling business created and added some small games to the App Store before, in 2010, launching Fruit Ninja—a game downloaded 1 billion+ times before its fifth birthday. Jetpack Joyride followed in 2011 and has grown to 750 million+ downloads.

“Fruit Ninja can fluctuate between 1 million and 1.5 million daily active users, while JetPack Joyride has between 500,000 and 900,000 daily active users,” says Rinal Deo, Chief Financial Officer and Product Manager at Halfbrick. “Another one of our games—Dan the Man—has around 300,000 daily active users.”

By early 2020, Halfbrick Studios had published more than 23 titles and complemented its mobile focus with virtual reality games available on platforms

such as Steam, Oculus, and PlayStation. The business has also ventured into developing games on Xbox. Halfbrick Studios employs about 35 people at its headquarters in Brisbane, Australia, plus another 11 in Spain and a small team in Bulgaria, while its licensing and entertainment business operates out of the United States.

In early July 2020, the team was preparing to launch version 2 of Fruit Ninja.

Halfbrick Studios began operations with in-house systems before electing to use a multinational cloud service to run a consolidated back end for many of its mobile games. However, the business wanted to reduce its infrastructure and system management load, lower its costs, and improve the agility and speed of its software development. These priorities extended to its analytics capabilities that Nik Kyriakidis, Lead Engineer, described as “hand-crafted” analysis of a large and fast-growing data lake incorporating data from its fast-growing Fruit Ninja and Jetpack Joyride games.

Cost optimization, reuse, and tooling drives move to Google Cloud

Halfbrick Studios reviewed its options and initially opted to migrate its analytics and mobile development to Google Cloud—with a view to moving all its workloads across in the longer term. “The decision came down to cost optimization, reuse, and tooling—specifically the integration of the Firestore (<https://firebase.google.com/>) mobile

development platform with Google Cloud,” says Kyriakidis. “Because we are mobile app focused, we see Firebase as key to reducing what we have to build. Then, having the power of Firebase backed by Google Cloud (<https://cloud.google.com/>) for additional capabilities is beneficial from a cost perspective and a developer perspective,” he says.

Halfbrick Studios uses Firebase as its mobile backend platform and Firestore (/firestore) to store player data, while a BigQuery (/bigquery) analytics data warehouse provides analytics data storage and reporting. The business also employs Cloud Run (/run) to deploy its custom in-house-developed APIs that cover server-side business logic, data migrations between different systems for different users and big data analytics ingress, with code recycled from its monolithic database service currently running in another cloud, packaged in Docker containers and deployed to the Google Cloud service. In the longer term, the business aims to decommission the workloads running in the service as part of the transition to Google Cloud.

The business migrated its analytics to Looker (/looker), a product Google acquired in 2019. “Looker is a very usable, effective data platform and myself and our product managers use it every day,” says Kyriakidis.

The business is running game-level A/B testing by modifying remote configuration for feature enablement on Firebase, enabling easy changes to values for push messaging and events. These changes may include directing certain users to an in-

game shop during an event and sending others to play the game to win particular items. “Also, the Cloud Firestore database allows us to sync data on the client in real time,” says Deo. “We now have configuration files on the server and the client—which we didn’t have on previous solutions—so we have the client as the source of truth and the functionality to modify it easily.”

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—Nik Kyriakidis, Lead
Engineer, Halfbrick
Studios

Cloud Run delivers greatest benefit

According to Kyriakidis, Cloud Run is the Google Cloud service of greatest benefit to the business, as it reduces operational costs as well as compute and provisioning. “In addition, because Cloud Run is completely stateless, our API development life cycle is now incredibly simple,” he says.

“When I started at Halfbrick Studios, we planned to hire more cloud services engineers to manage infrastructure so I could focus on adding functionality to our games. Cloud Run eliminated the need to add another engineer because we did not need to manage infrastructure or instances—it is just ‘set and forget.’”

Halfbrick Studios is now building on its initial investments by using machine learning (ML) to provide a more intuitive, personalized experience for

gamers. “We regard machine learning as the next frontier in gaming,” says Deo. We need to push into that area and see what we can accomplish.” The business aims to use ML to recognize player behavior and provide a fun, personalized playing experience complemented by clever monetization.

The organization plans to focus its ML activities on BigQuery ML and its API, with Kyriakidis citing the need to move data across from another cloud service so the business can model the propensity of its users. “Other cloud providers assume you have deep expertise in ML—we don’t have that in our office, but Google Cloud’s ML services are highly intuitive and suit our cloud programmers,” says Kyriakidis.

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*—Rinal Deo, Chief
Financial Officer and
Product Manager,
Halfbrick*

Liveops enhances engagement

Halfbrick Studios continues to evolve its products and Google Cloud has played a key role in the transition to the current “free to play with ad monetization” model. “Our focus over the last two to three years has been on free-to-play games with ‘liveops’—making changes such as releasing offers or adjusting prices on the fly,” says Kyriakidis. “Features within Firebase allow us to control data that tweaks the experience in our games without having to release new builds for every incremental change.

“As a result, team members not highly skilled in coding or analytics can check information about player behavior and adapt games by changing variables in the code to, say, change difficulty at certain levels or offer different rewards to players.”

The business views liveops as key to the success of forthcoming games such as the highly anticipated Fruit Ninja 2. “Without liveops enabled by Google Cloud, we cannot make our games more engaging, more customized, and more personal—which is one of our key business objectives,” says Deo.

Meeting user retention metrics

Liveops enabled by Google Cloud is also the cornerstone of Halfbrick Studios's efforts to achieve game metrics based on long-term player retention.

"We focus not only on keeping players on day one, but on day 30, day 90, and day 180," says Deo. "With liveops, we can provide enough content to motivate players to stay for the longer term."

"We strive for 45% of users who install games to return to the game that same day—in line with top-performing games, about 10% 30 days after installation, and 5% at day 60 and day 180."

Using platform as a service through Google Cloud enables Halfbrick Studios to push new games, updates and features to market faster than when using more manual systems. "We can capitalize on a segment of the game market in a specific genre by providing a new feature or by creating a particular style of game," says Deo. "Having pre-built features allows us to be quicker to market and potentially consume a percentage of the revenue available."

He cites as an example the rebuild of the game *Monster Dash* to soft-launch status within a month—the fastest launch completed by the business in the last 12 months. "Leveraging our previous technology, we would have needed three months to reach production soft launch," explains Kyriakidis.

"Testing and sending products live: that quickly gives us a lot more validation—we're not dedicating resources for three months to see whether a game works or not," says Deo. "That is a huge benefit and aligns with one of our core values, which is

experimenting and pushing products and features out quickly to test the market.”

“Overall, Google Cloud runs a very granular level of scalability, and performance is far above every other cloud service that provides compute. There is no comparable product in terms of price, scale, and power.”

—Nik Kyriakidis, Lead Engineer, Halfbrick Studios

Lowering costs from \$24 per day to \$4 per day

The business also found Google Cloud was considerably cheaper than its incumbent provider when sending analytics and event data to Cloud Run and BigQuery—even when that data resided in that provider's cloud.

“We ran an analysis to determine the lowest cost for running a sample size of data on an instance in the other cloud service, versus exporting analytics and event data from that service to Cloud Run and BigQuery,” explains Kyriakidis. “It cost us AUD\$24 per day to run through the alternative cloud service, versus AUD\$4 per day to run it through Google Cloud—including sending data from one cloud to another and back again.”

The data exporting is required for integration with the Looker platform. “The alternative to our Cloud Run architecture was run Cloud Functions to read data from blob storage in another cloud service into a Looker-compatible database,” says Kyriakidis. “We expect the advent of BigQuery Omni—which will allow us to access and analyze data across multiple clouds from the BigQuery interface—to further reduce our costs.”

He also attributed part of the benefit to the fact Cloud Run could serve a wider range of requests per instance than alternative solutions—enabling the business to control scaling while maintaining high throughput.

Halfbrick Studios views Google Cloud and Firebase as integral to its longer term success. “Overall, Google Cloud runs a very granular level of scalability,

and performance is far above every other cloud service that provides compute,” Kyriakidis concludes. “There is no comparable product in terms of price, scale, and power.”